



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru - 560064



ITDA Audit

AN INTERNSHIP REPORT

Submitted by

TANUSHREE R - 20221CBD0029

Under the guidance of,

Ms. SANDHYA L

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND TECHNOLOGY (BIG DATA)



PRESIDENCY UNIVERSITY

BENGALURU

APRIL 2026



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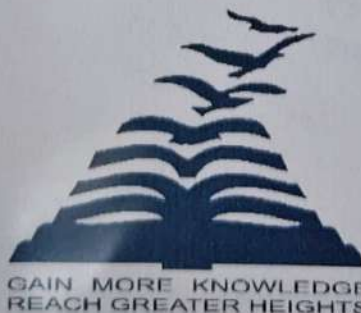
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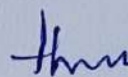
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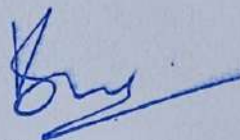
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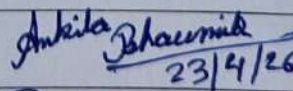
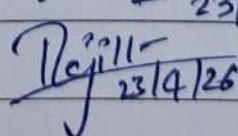
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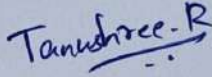
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DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND TECHNOLOGY (BIG DATA), at Presidency University, Bengaluru, named **Tanushree R** hereby declare that the internship work titled **"ITDA Audit"** has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE and TECHNOLOGY (Spl. Division) during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Tanushree R

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DATE: 23rd April 2026

INTERNSHIP COMPLETION CERTIFICATE



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December 23, 2025

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Dear Tanushree R

Placement as Intern

With reference to your application for a project work assignment as a part of your graduation study, we are pleased to inform you that you are being allotted a project work assignment in the **IT, Data & Analytics Department of Audit & Assurance** Function of our Organization in **Bengaluru** location from **January 5, 2026 to July 3, 2026** on the following terms and conditions:

1. You will be assigned a Manager under whose supervision you will work on the project assigned to you.
2. You will be paid a monthly stipend as described in Annexure A. Please note that the tax will be deducted on the stipend amount.
3. You shall complete your project in accordance with the requirements and guidance of the Manager of the Organization, and maintain qualitative standards as required.
4. Notwithstanding the fact that your engagement is internship, for your study purpose. It is expected of you to maintain the discipline, dignity, honor and goodwill of the Organization.
5. The arrangement is not that of an employer and an employee and as such you shall not be eligible to any allowances or other benefits as may be available to the employees of the Organization.
6. You will observe the rules & regulations and discipline of our office, and also maintain complete confidentiality and secrecy of the matters pertaining to our Office and/or any data that has been provided to you in the course of your project work. You will be permitted to attend any classes in the college at the discretion of the Manager, if so called for during the period of your project assignment.

ACKNOWLEDGEMENTS

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the internship.

I would like to sincerely thank my internal guide **Ms. Sandhya L**, Assistant Professor, Presidency School of Computer Science and Engineering (Spl. Division) Presidency University, for her moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

I am also thankful to **Dr. S. Pravinthraja**, Professor, Head of the Department, Presidency School of Computer Science and Engineering (Spl. Division) Presidency University, for his mentorship and encouragement.

I express my cordial thanks to **Dr. Shakkeera L**, Associate Dean, Presidency School of computer Science and Engineering (Spl. Division), Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

I am grateful to **Dr. Geetha A, PSCS**, (Spl. Division), School Internship Coordinator and **Mrs. Nisha Robin Rohit, PSCS** (Spl. Division) **Program Project Coordinator**, Presidency School of Computer Science and Engineering, for facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

I am also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

Tanushree R

ABSTRACT

This internship was conducted within the **Information Technology, Data & Analytics (ITDA)** Capability & Enhancement Center (CEC) at **Deloitte**, focusing on the specialized domain of **IT Audit**. The primary background of this project involves performing **Information Technology General Controls (ITGC)** testing to ensure the integrity, security, and reliability of system-generated data that supports financial reporting. As organizations increasingly rely on complex digital infrastructures, the audit process must move beyond traditional financial statements to rigorously evaluate the underlying systems and applications, specifically focusing on Access Security, Change Management, and Data Center operations. The approach utilized during this internship centered on the four stages of the **Audit Lifecycle**. The core methodology involved two testing phases. Technical execution was driven by data-driven audit analytics using the **Deloitte Omnia** ecosystem and the **Automated Control Testing Tool (ACTT)**. These tools were employed to perform automated testing across diverse enterprise environments, including **SAP, HANA, and LINUX**, with a specific focus on validating user access privileges and system settings. Beyond core testing, the project involved rigorous internal quality measures, such as the **DQA Review** roadmap, to ensure documentation accuracy and archival readiness. This process required the preparation of detailed **Rule Templates** in Excel to define server details, system names, and versions before processing raw client data through automated tools. For broader engagement management, the internship also integrated **ITS Support** to track project progress via the **ITS Tracker** website and maintain communication with various Location Managers and Single Points of Contact (SPOCs). The results of this internship included the successful execution of comprehensive audit procedures, such as **User Access Management (UAM)** reviews, **Change Management (CM)** sampling, and **Role Change** analysis using advanced Excel-based logic and methodologies. Through the preparation of **Request for Information (RFI)** documents and **2342S** control deficiency reports, the project facilitated the identification of inappropriate access privileges such as unauthorized table access or improper segregation of duties and ensured adherence to industry-standard security policies. Ultimately, the internship culminated in the documentation of formal **Workpapers**, providing the necessary evidence and justification to validate the control environment for the client's enterprise systems.

TABLE OF CONTENTS

Sl. No.	Title	Page No.
	Declaration	iii
	Acknowledgement	v
	Abstract	vi
	List of Figures	ix
	List of Tables	x
	Abbreviations	xi
1.	Introduction 1.1 Background 1.2 Statistics of Internship 1.3 Prior existing technologies 1.4 Proposed approach 1.5 Objectives 1.6 SDGs 1.7 Overview of internship report	1 - 5
2.	Literature Review	6 - 9
3.	Methodology	10 - 12
4.	Project Management 4.1 Internship timeline 4.2 Risk analysis 4.3 Internship budget	13 - 18
5.	Analysis and Design 5.1 Requirements 5.2 Block Diagram 5.3 System Flow Chart 5.4 Choosing devices 5.5 Designing units 5.6 Standards 5.7 Mapping with IoTWF reference model layers 5.8 Domain model specification 5.9 Communication model 5.10 IoT deployment level	19 - 28

TABLE OF CONTENTS

Sl. No.	Title	Page No.
	Declaration	iii
	Acknowledgement	v
	Abstract	vi
	List of Figures	ix
	List of Tables	x
	Abbreviations	xi
1.	Introduction 1.1 Background 1.2 Statistics of Internship 1.3 Prior existing technologies 1.4 Proposed approach 1.5 Objectives 1.6 SDGs 1.7 Overview of internship report	1 - 5
2.	Literature Review	6 - 9
3.	Methodology	10 - 12
4.	Project Management 4.1 Internship timeline 4.2 Risk analysis 4.3 Internship budget	13 - 18
5.	Analysis and Design 5.1 Requirements 5.2 Block Diagram 5.3 System Flow Chart 5.4 Choosing devices 5.5 Designing units 5.6 Standards 5.7 Mapping with IoTWF reference model layers 5.8 Domain model specification 5.9 Communication model 5.10 IoT deployment level	19 - 28

	5.11 Functional view 5.12 Mapping IoT deployment level with functional view 5.13 Operational view 5.14 Other Design	
6.	Hardware, Software and Simulation 6.1 Hardware 6.2 Software development tools 6.3 Software code 6.4 Simulation	29 - 31
7.	Evaluation and Results 7.1 Test points 7.2 Test plan 7.3 Test result 7.4 Insights	32 - 33
8.	Social, Legal, Ethical, Sustainability and Safety Aspects 8.1 Social aspects 8.2 Legal aspects 8.3 Ethical aspects 8.4 Sustainability aspects 8.5 Safety aspects	35 - 37
9.	Conclusion	38
	References	40
	Base Paper	41
	Appendix	43

LIST OF FIGURES

Sl. No.	Table Name	Table Caption	Page No.
1	Figure 1.1	Sustainable development goals	4
2	Figure 3.1	The V-Model Methodology	10
3	Figure 4.1	Gantt Chart for Internship Project Timeline	13
4	Figure 5.2.2	Functional Block Diagram of the IT Audit System	21
5	Figure 5.3	System Flowchart for ITGC Testing	22
6	Figure 5.8	Request-Response Communication Model for IT Audit	27
7	Figure 5.9	Functional View of the IT Audit System	28

LIST OF FIGURES

Sl. No.	Table Name	Table Caption	Page No.
1	Figure 1.1	Sustainable development goals	4
2	Figure 3.1	The V-Model Methodology	10
3	Figure 4.1	Gantt Chart for Internship Project Timeline	13
4	Figure 5.2.2	Functional Block Diagram of the IT Audit System	21
5	Figure 5.3	System Flowchart for ITGC Testing	22
6	Figure 5.8	Request-Response Communication Model for IT Audit	27
7	Figure 5.9	Functional View of the IT Audit System	28

LIST OF TABLES

Sl. No.	Table Name	Table Caption	Page No.
1	Table 2.1	Summary of Literature Reviews	8
2	Table 4.1	Internship Project Timeline	13
3	Table 4.2.1	PESTLE Analysis for Deloitte IT Audit Engagement	15
4	Table 4.2.2	Project Risk Assessment Matrix	16
5	Table 4.3.2	Engagement Resource Budget	18
6	Table 5.1	Summarizing Audit System Requirements	19
7	Table 7.3	Observations of Sampling and Ingestion Units	33

ABBREVIATIONS

Abbreviation	Full Form
ACTT	Automated Control Testing Tool
CEC	Capability & Enhancement Center
CM	Change Management
DCNO	Data Center and Network Operations
DQA	Data Quality Assurance
GITC	General IT Controls
GRC	Governance, Risk, and Compliance
ITDA	Information Technology, Data & Analytics
ITGC	Information Technology General Controls
RAIT & RAWC	Risk Associated with IT & Risk Associated with Control
RFI	Request for Information
SoD	Segregation of Duties
SPOC	Single Point of Contact
UAM	User Access Management

CHAPTER 1

INTRODUCTION

1.1 Background

The digital transformation of global enterprises has shifted the landscape of corporate governance, making the integrity of Information Technology (IT) systems a cornerstone of financial reliability. Within the domain of **Audit & Assurance**, **IT Audit** serves as a specialized function responsible for evaluating system-generated data and the underlying digital infrastructure rather than traditional financial statements. This project was conducted within the **Information Technology, Data & Analytics (ITDA)** Capability & Enhancement Center (CEC) at **Deloitte**.

The core focus of this internship involves the execution of **Information Technology General Controls (ITGC)**, which are the foundational policies and procedures ensuring that information systems are managed in a structured and controlled manner. These controls are categorized into three primary domains: **Access Security**, **System Change Control (Change Management)**, and **Data Center and Network Operations (DCNO)**. By validating these controls through a structured **Audit Lifecycle**: comprising Planning, Fieldwork, Concluding, and Archiving the project ensures that only authorized individuals have access to critical systems and that all modifications to the digital environment are documented and approved.

1.2 Statistics

The necessity for rigorous IT auditing is driven by the increasing complexity of enterprise resource planning (ERP) systems like **SAP**, **HANA**, and **LINUX**. In the current regulatory environment, **General IT Controls (GITC)** are essential for maintaining compliance and mitigating the **Risk Associated with IT (RAIT)** and the **Risk Associated with Control (RAWC)**.

The need for this project is highlighted by the scale of modern enterprise data; for instance, in **User Access Management (UAM)** testing, populations often exceed 365 entries,

requiring specific sampling frequencies (ranging from 25 to 35 samples) to ensure statistical significance in the audit findings. Regionally, as a global leader in audit services, Deloitte utilizes the **Omnia** platform to standardize and digitize these data-driven processes. This centralized approach addresses the need for real-time transparency and enhanced collaboration between specialist teams and auditors, ensuring that audit diagnostics are monitored constantly to prevent **Segregation of Duties (SOD)** conflicts and unauthorized system access.

1.3 Prior Existing Technologies

Before the integration of advanced, cloud-based audit platforms, IT auditing relied heavily on manual documentation and non-structured MS Office files. Prior methodologies often involved fragmented workflows where data was imported or emailed across platforms, leading to limited visibility into the audit engagement file.

Key technologies and tools previously used in the audit process include:

- **Manual Excel Testing:** Historically, auditors manually performed logic such as VLOOKUPS and concatenation without the benefit of centralized, automated tools to identify discrepancies in user access or change management logs.
- **Traditional SAP/LINUX Logs:** Auditors would manually extract and review T-codes and logs such as **SM50**, **SM36**, or **SM37** to track job scheduling and system changes.
- **Non-Integrated Risk Assessments:** Risk assessments and IT specialist discussion questionnaires were often maintained as separate, non-structured documents rather than being part of an integrated, real-time project plan.
- **Localized Tracking:** Internal project tracking was frequently done via fragmented trackers rather than centralized web-based tools like the **ITS Tracker** website currently used to monitor manager updates and project milestones.

1.4 Proposed Approach

- **Aim of Project:** The primary aim of this project is to evaluate and validate **Information Technology General Controls (ITGC)** across complex enterprise environments, including

SAP, HANA, and LINUX, to ensure the integrity, security, and reliability of system-generated data supporting financial reporting.

- **Motivation:** This project is motivated by the critical need for organizations to mitigate the **Risk Associated with IT (RAIT)** and the **Risk Associated with Control (RAWC)**. By identifying control deficiencies through rigorous testing, organizations can prevent unauthorized system modifications and ensure regulatory compliance.
- **Proposed Approach:** The methodology utilizes a data-driven audit analytics approach powered by the **Deloitte Omnia** ecosystem and the **Automated Control Testing Tool (ACTT)**. The project follows a structured four-stage **Audit Lifecycle**: Planning, Fieldwork, Concluding, and Archiving incorporating **Design and Implementation (D&I)** testing followed by **Operating Effectiveness (OE)** testing to ensure continuous control performance.
- **Applications of the Project:** The project is applied to enterprise-level security and governance, specifically in managing **User Access Management (UAM)**, **System Change Control**, and **Data Center and Network Operations (DCNO)**. It enables the systematic identification of inappropriate access through the preparation of **Request for Information (RFI)** and **2342S** deficiency reports.
- **Limitation of the Proposed Approach:** A significant limitation is the heavy reliance on the quality, completeness, and accuracy of raw data provided by the client. Furthermore, while tools like **ACTT** automate the processing, high-level **DQA Reviews** and the interpretation of complex business justifications still require manual expert intervention.

1.5 Objectives

1. **System Management:** To standardize and digitize the auditing process using the **Deloitte Omnia** cloud-based platform, ensuring centralized documentation and real-time monitoring of engagement diagnostics.
2. **Security:** To validate **Access Security** by testing user provisioning, deprovisioning, and **Segregation of Duties (SoD)** across **SAP, HANA, and LINUX** platforms to ensure only authorized individuals have system access.
3. **Analysis:** To perform quantitative sampling and analysis for **UAM** and **Change Management (CM)** by applying advanced Excel logic (including **CONCATENATE**, **VLOOKUP**, and **MOD** functions) to identify discrepancies in large data populations.

4. **Behavior:** To evaluate system behaviors and job performance, such as failed job monitoring and unauthorized program executions through the detailed review of **SAP** and **UNIX** system logs.
5. **Deployment:** To execute a structured **DQA Review** roadmap and **ITS Support** tracker, ensuring that all audit workpapers and evidence are accurately documented, reviewed, and archived within the specified engagement timeline.

1.6 Sustainable Development Goals (SDGs)

The objectives of this IT Audit project align with several **United Nations Sustainable Development Goals (SDGs)**:



Figure 1.1 Sustainable development goals

- **SDG 8: Decent Work and Economic Growth:** By strengthening IT governance and internal controls, the project promotes institutional stability and protects the digital infrastructure essential for sustained economic growth.
- **SDG 9: Industry, Innovation, and Infrastructure:** The implementation of innovative, data-driven platforms like **Omnia** and **ACTT** demonstrates a commitment to upgrading technological capabilities and ensuring reliable, secure industrial infrastructures.
- **SDG 16: Peace, Justice, and Strong Institutions:** The focus on **Audit & Assurance** and internal ITGC testing promotes accountability and transparency within large organizations, contributing to the development of effective and accountable institutions.

1.7 Overview of project report

This report documents the automation of IT Audit processes within Deloitte's ITDA CEC. Chapter 1 introduces the project's focus on Information Technology General Controls (ITGC), while Chapter 2 reviews literature on data-driven audit workflows. Chapter 3 details the V-Model methodology employed for control testing. Chapter 4 outlines project management, including risk analysis and resource budgeting. Chapters 5 and 6 describe the system design, hardware infrastructure, and software tools like Deloitte Omnia and ACTT. Chapter 7 presents evaluation results, including sampling methodologies and deficiency analysis. Chapter 8 discusses the project's broader social, legal, ethical, and sustainability implications. Finally, Chapter 9 summarizes the project's achievements in audit automation and provides recommendations for future enhancements.

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Existing Literature

Reference Paper 1: Oko-Odion, C. and Udoh, O.R., 2024. Leveraging technology in internal audit processes for streamlined management and risk oversight. *International Journal of Science and Research Archive*, 13(2), pp.3077-3100. [1]

Recent academic journals emphasize the transformation of the auditing profession through the introduction of data analytics and automated workflows. Research suggests that while traditional audits relied on manual representative sampling, modern AI-enabled platforms are now applied at various stages, from risk identification in the planning phase to building real-time dashboards for control deviations. The literature identifies that automated internal control testing improves detection rates by 20–70% compared to manual methods. This research directly supports your use of the **Deloitte Omnia** platform to standardize and digitize the audit process for greater visibility and collaboration.

Reference Paper 2: El Arass, M., Tikito, I. and Souissi, N., 2018, June. An audit framework for data lifecycles in a big data context. In *2018 International Conference on Selected Topics in Mobile and Wireless Networking (MoWNeT)* (pp. 1-5). IEEE. [2]

Scholars have developed end-to-end models that drive the audit lifecycle through a systematic roadmap of planning, execution (fieldwork), and reporting. This research advocates for a risk-based approach to navigate the complexities of digital infrastructures, ensuring that audit objectives align with organizational goals. The study highlights that the execution phase must leverage technology for extensive data collection and advanced analytics. This aligns perfectly with your experience in the **four-stage Audit Lifecycle** at Deloitte: Planning, Fieldwork, Concluding, and Archiving, where automated tools are used to maintain consistency.

Reference Paper 3: Chuprunov, M., Bachmann, R. and Nowatzki, A., 2013. *Auditing and GRC Automation in SAP* (pp. I-XXXII). Berlin: Springer. [3]

Research into SAP security guidelines emphasizes the use of automated tools to identify vulnerabilities and deviations from standards in critical authorizations and Segregation of Duties (SoD). The literature describes a "Quick Audit" procedure involving check policy definition, job scheduling for analysis, and automated validation of security-relevant SAP parameters. Findings indicate that automated suites can save approximately 20 working days per audit, significantly relieving the burden on IT departments. This supports your technical work in **SAP Testing**, where you prepare rule templates and use **ACTT** to process raw client data for control validation.

Reference Paper 4: Chuprunov, M., Bachmann, R. and Nowatzki, A., 2013. *Auditing and GRC Automation in SAP* (pp. I-XXXII). Berlin: Springer. [4]

Conference papers focusing on SAP Governance, Risk, and Compliance (GRC) discuss the modernization of Identity and Access Management (IAM) through automation. Research reports that automating user termination and access reviews can increase task identification speed by 35% and overall compliance effectiveness by 50%. The study demonstrates that real-time enforcement of the user lifecycle minimizes manual intervention and prevents unauthorized access by automatically disabling inactive accounts. This validates your testing of **Access Security** controls (SAP.01–SAP.03) and the importance of timely user deprovisioning.

Reference Paper 5: Huang, F., No, W.G., Vasarhelyi, M.A. and Yan, Z., 2022. Audit data analytics, machine learning, and full population testing. *The Journal of Finance and Data Science*, 8, pp.138-144. [5]

As audit demand grows in the era of big data, researchers are proposing a shift from manual sampling to "full population auditing" based on machine learning. This method addresses the risks arising from traditional sampling by extending the audit scope to all data points through self-learning features. While the project currently utilizes **advanced Excel sampling (MOD/VLOOKUP)** for large populations, the literature suggests that moving toward full population testing is the next evolutionary step for internal control audits. This research justifies your focus on sampling populations exceeding 365 entries to maintain statistical significance.

Reference Paper 6: Sarferaz, S., 2025. *Implementing Agentic AI Into ERP*. Presidency School of Computer Science and Engineering, Presidency University.

Software. *IEEE Access*. [6]

Recent IEEE conference papers explore the implementation of agentic AI systems within Enterprise Resource Planning (ERP) software to automate troubleshooting and incident management. These systems convert management guides into executable plans with minimal human intervention. While the project focuses on **Linux and HANA testing**, this literature highlights the broader industry trend of using executable scripts (similar to your **ACTT scripts**) to extract data and assess impacts in non-production environments before final auditing.

2.2 Summary of Literature Review

Table 2.1: Summary of Literature Reviews

Sl. No.	Article Title, Published year, Journal/Conference name	Methods	Key Features	Merits	Demerits
1	Leveraging technology in internal audit processes..., 2024, Int. Journal of Science and Research Archive	Data Analytics and Automated Workflows	AI-enabled platforms applied across planning and risk identification stages.	Improves detection rates of control deviations by 20–70% over manual methods.	Real-time dashboards require high-quality, continuous data streams to be effective.
2	An Audit Framework for Data Lifecycles in a Big Data Context, 2018, IEEE (MoWNeT)	Risk-based End-to-End Audit Model	Systematic roadmap covering Planning, Fieldwork, Concluding, and Archiving.	Aligns audit objectives strictly with organizational goals and digital infrastructure.	Complexity of digital infrastructures can make the execution phase resource-intensive.
3	Auditing and GRC Automation	"Quick Audit"	Automated validation of	Saves approximately 20	Automated suites require

	in SAP, 2013, Springer	Procedure & Job Scheduling	security-relevant SAP parameters and SoD rules.	working days per audit; reduces burden on IT staff.	precise initial check policy definitions to avoid false positives.
4	Auditing and GRC Automation in SAP (IAM focus), 2013, Springer	Automated Identity & Access Management (IAM)	Real-time enforcement of user lifecycle and automated termination.	Increases task identification speed by 35% and compliance by 50%.	Minimal manual intervention can lead to oversight if the automation logic is not updated.
5	Audit data analytics, machine learning..., 2022, The Journal of Finance and Data Science	Full Population Auditing via Machine Learning	Extension of audit scope to all data points using self-learning features.	Addresses inherent risks of traditional manual sampling in big data environments.	Requires significant computational power and specialized ML knowledge for implementation.
6	Implementing Agentic AI Into ERP Software, 2025, IEEE Access	Agentic AI Systems & Executable Scripts	Conversion of management guides into executable troubleshooting plans.	Minimizes human intervention in complex ERP cloud environments and incident management.	High complexity in converting qualitative management guides into accurate automated scripts.

CHAPTER 3

METHODOLOGY

For the execution of the IT Audit project at Deloitte, the **V-Model (Validation and Verification Model)** is implemented to ensure that every control requirement is systematically designed and rigorously tested. Unlike a standard linear model, the V-Model provides a dual-track approach: the **Verification Phase** ensures that the audit plan and control designs meet specialized standards, while the **Validation Phase** confirms that these controls operate effectively within the client's live environment.

3.1 The V-Model in the Audit Lifecycle

As illustrated in **Figure 3.1**, the V-Model demonstrates a direct relationship between development phases and their corresponding testing stages. In the context of an IT Audit, this ensures that the risk identified during planning is directly addressed during fieldwork and final reporting.

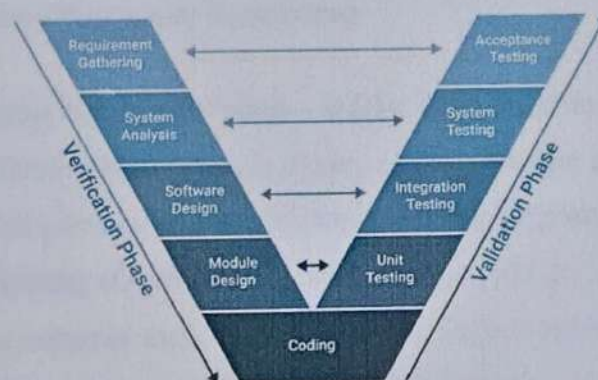


Figure 3.1: The V-Model Methodology

The verification stage includes phases such as Requirement Gathering and System Analysis, whereas the validation phase includes unit, system, and acceptance testing. The following sections map the specific internship tasks to these phases.

3.2 Mapping Internship Stages to the V-Model

The audit process is divided into two main streams of work that correspond to the "V"

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structure:

A. The Verification Phase (Design & Planning)

1. **Requirement Gathering (Planning):** In this initial stage, the audit scope is defined by identifying the **Risk Associated with IT (RAIT)** and **Risk Associated with Control (RAWC)**. We determine the specific systems to be audited, such as **SAP, HANA, or LINUX**.
2. **System Analysis (Control Definition):** The high-level audit requirements are broken down into the three primary domains: Access Security, Change Management, and Data Center Operations.
3. **Software/Module Design (Rule Template Preparation):** For the technical execution, **Rule Templates** are designed in Excel. These templates define the server details (system, name, version) and the specific logic that the **Automated Control Testing Tool (ACTT)** will use to process raw data.
4. **Coding (Script Execution):** This represents the "bottom of the V," where raw client files are processed through **ACTT** to generate the populated files needed for testing.

B. The Validation Phase (Testing & Execution)

1. **Unit Testing (Design and Implementation - D&I):** Each specific control (e.g., SAP.01 or UNIX.05a) undergoes **D&I testing**. In this phase, a single sample is taken to verify that the control is designed and implemented correctly by adding specific evidence.
2. **System/Integration Testing (Operating Effectiveness - OE):** This phase tests the full population to ensure the controls worked consistently over the entire audit period. Advanced Excel logic including **VLOOKUP, MOD, and CONCATENATE** is used to sample and analyze **User Access Management (UAM)** and **Change Management (CM)** data.
3. **Acceptance Testing (DQA Review & Concluding):** The final stage involves an internal **DQA Review (Data Quality Assurance)**. We resolve alerts, verify audit memos, and perform Power BI runs to ensure the Workpapers are accurate and ready for **Archiving**.

3.3 Methodology Tools

The V-Model is supported by specialized tools within the Deloitte ecosystem:

- **Deloitte Omnia:** The centralized cloud platform used for real-time monitoring and integrated project planning.
- **ACTT:** Used for automated extraction and processing of system-generated data.
- **D-connect:** A secure portal for sharing documents, **Request for Information (RFI)**, and 2342S deficiency reports with the client.
- **ITS Tracker:** A web-based tool used to support project management by tracking manager updates and location-specific SPOCs.

CHAPTER 4

PROJECT MANAGEMENT

4.1 Internship Timeline

To ensure the successful execution of the IT Audit within the established engagement deadlines, a structured project timeline is essential. This project utilizes a **Gantt Chart** approach to provide a visual roadmap, outlining tasks, their durations, and critical milestones in chronological order.

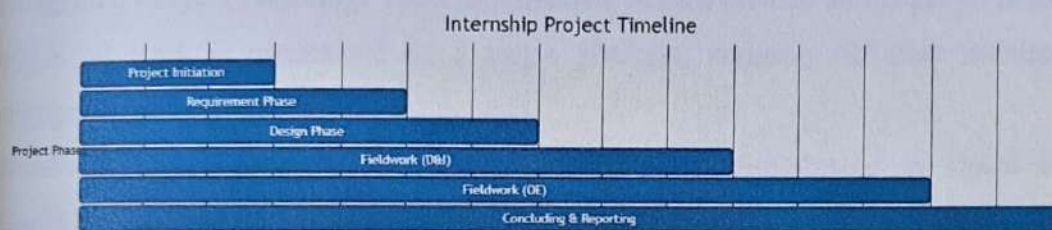


Figure 4.1: Gantt Chart for Internship Project Timeline

Table 4.1: Internship Project Timeline

Sl. No.	Major Task	Start Week	End Week	Milestone/Deadline
1	Project Initiation: Selection of IT Audit domain and training	Week 1	Week 3	Training Completion
2	Requirement Phase: Identifying RAIT/RAWC and scope	Week 4	Week 5	Scope Finalization
3	Design Phase: Preparing Rule Templates for SAP, HANA, & Linux	Week 6	Week 7	Rule Template Approval
4	Fieldwork (D&I): Automated testing via ACTT & initial sampling	Week 8	Week 10	D&I Documentation
5	Fieldwork (OE): Full population testing & RFI/2342S generation	Week 11	Week 13	RFI Response Receipt

6	Concluding & Reporting: DQA Review, Archiving, and Final Report	Week 14	Week 15	Project Archival
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As summarized in **Table 4.1**, the timeline focuses on a phased approach starting from training to the final archival. The critical milestones include the completion of technical training by Week 3 and the submission of the final Workpapers and report by Week 15.

4.1.2 Suitability for the Project

The use of a structured timeline and Gantt chart is highly suitable for an IT Audit internship at Deloitte for several reasons:

- **Integrated Project Planning:** Tools like **Deloitte Omnia** provide an integrated project plan where all work is maintained on a single platform, ensuring real-time monitoring of engagement diagnostics.
- **Deadline Management:** Audit engagements are strictly time-bound. A visual schedule allows for tracking the status of **Request for Information (RFI)** IDs and ensures manager updates are recorded on time via the **ITS Tracker**.
- **Resource Allocation:** By identifying tasks like **SAP Testing**, **Linux Testing**, and **DQA Reviews** in a chronological sequence, the timeline helps in coordinating with SPOCs and location managers to prevent bottlenecks in evidence collection.
- **Quality Assurance:** The final weeks are dedicated to the **DQA Roadmap**, which requires unlocking the file for internal review and resolving alerts before the hard deadline for archiving.

4.2 Risk Analysis

To ensure the successful delivery of the IT Audit engagement, it is critical to perform a comprehensive risk assessment. This project utilizes the **PESTLE analysis** framework to evaluate external factors that could impact the audit lifecycle and the **Risk Matrix** to prioritize mitigation efforts.

4.2.1 PESTLE Analysis

The PESTLE analysis assesses how macro-environmental factors influence the project's

success, allowing for proactive risk mitigation.

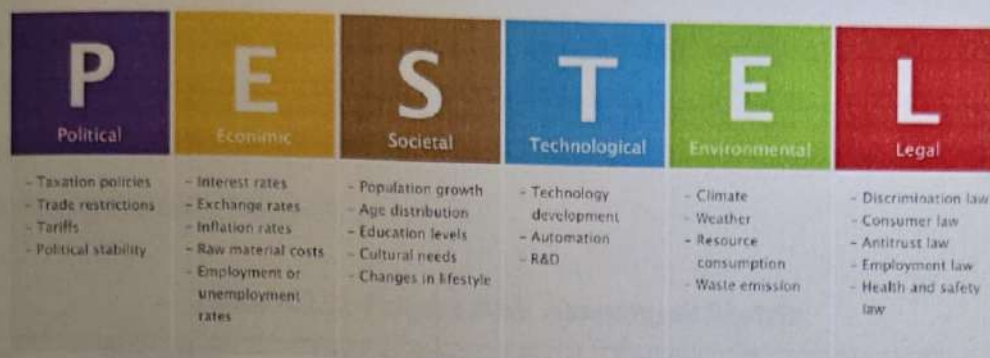


Table 4.2.1: PESTLE Analysis for Deloitte IT Audit Engagement

Factor	Description of Risk	Mitigation Strategy
Political	Changes in government data privacy regulations (e.g., updates to IT acts) impacting audit standards.	Regular consultation with Deloitte's legal and policy teams to ensure compliance with the latest mandates.
Economic	Budgetary constraints on the client side leading to delays in providing necessary system access or logs.	Early communication of resource requirements and maintaining a clear ITS Tracker to flag delays to managers.
Societal	Shift toward remote work affecting the availability of client SPOCs for evidence gathering and interviews.	Utilizing D-connect for secure, asynchronous document sharing and scheduling virtual walkthroughs.
Technological	System downtime in client environments (SAP, HANA, LINUX) preventing raw data extraction for ACTT.	Planning "Fieldwork" during low-activity windows and utilizing Rule Templates to ensure data is processed quickly once received.
Legal	Non-compliance with international audit standards or failure to maintain Segregation of Duties (SoD) during testing.	Strict adherence to the DQA Review roadmap and resolving alerts before final project archival.
Environmental	Regional infrastructure issues (e.g., power outages in Bengaluru) affecting connectivity to the Omnia cloud platform.	Leveraging Deloitte's cloud-based infrastructure which allows for co-editing and real-time syncing across global locations.

4.2.2 Risk Assessment Matrix

To effectively manage the risks identified in the PESTLE analysis, a **Risk Assessment Matrix** is utilized to quantify and prioritize each risk based on its **Potential Consequence** and **Probability of Occurrence**. This ensures that high-impact risks receive immediate attention during the audit lifecycle.

Table 4.2.2: Project Risk Assessment Matrix

Risk Identifier	Potential Consequence (1-5)	Probability (1-5)	Risk Score	Risk Category	Mitigation Strategy
Data Integrity Breach (Inappropriate SAP/Linux access)	5 (Catastrophic)	2 (Unlikely)	10	Significant	Strict enforcement of Access Security controls and immediate flagging of unauthorized table access (SAP.07).
System Downtime (Client raw data unavailable)	4 (Major)	3 (Could Occur)	12	Significant	Proactive communication via the ITS Tracker and early requests for data through the D-connect portal.
Inaccurate Testing (Incorrect Rule Template logic)	3 (Moderate)	3 (Could Occur)	9	Medium	Rigorous DQA Review roadmap and secondary validation of Excel formulas (VLOOKUP, MOD).
Project Delay (Delayed RFI responses)	2 (Minor)	4 (Known to Occur)	8	Medium	Regular follow-ups with SPOCs and location managers; utilizing the ITS Support tracker for reminders.
Documentation Errors	2 (Minor)	2 (Unlikely)	4	Low	Standardizing documentation using

(Workpaper inconsistencies)					Deloitte structured items.	Omnia's built-in work items.
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As illustrated in **Table 4.2.2**, the risks are scored by multiplying the consequence rating with the probability rating. This methodology allows the team to focus on "Significant" and "High" risk items, such as system failures or operational downtime, that could result in the loss of critical data integrity or the failure to meet archival deadlines.

- **Significant/High Risk (Score 10-25):** Requires immediate management action and high-level internal guidance.
- **Medium Risk (Score 5-9):** Managed through routine audit procedures, such as **D&I** and OE testing.
- **Low Risk (Score 1-4):** Addressed through standard quality checks and automated diagnostics within the **Omnia** platform.

4.3 Internship Budget

The project budget for this IT Audit engagement focuses on the strategic allocation of enterprise resources rather than personal expenditures. Since the project is conducted within **Deloitte**, the "budget" represents the high-value infrastructure and professional man-hours invested by the firm to ensure audit quality and compliance.

4.3.1 Budgeting Steps

The budget for the audit lifecycle was established following these critical steps:

- **Step 1: List All Tasks and Resource Requirements:** All audit phases: Planning, Fieldwork, Concluding, and Archiving were mapped to identify necessary tools like **Deloitte Omnia**, **ACTT**, and secure portals like **D-connect**.
- **Step 2: Check Team Availability:** Coordination was performed with **Location Managers** and **SPOCs** to ensure specialist availability for testing **SAP**, **HANA**, and **LINUX** environments.

- **Step 3: Estimate Task Duration:** Based on the complexity of populations (e.g., UAM populations exceeding 365 entries), task durations were estimated to align with the ITS Tracker deadlines.
- **Step 4: Use Your Experience and Data:** Historical data from similar engagements and the Rule Templates were used to forecast the time required for automated data processing via ACTT.
- **Step 5: Set the Project Budget:** The final "budget" was set as a commitment of company-provided infrastructure and internal labor hours for **DQA Reviews**.
- **Step 6: Keep Track of the Project Budget and Assess the Team:** Real-time monitoring of engagement diagnostics via **Omnia** ensured that resource usage remained within the planned limits throughout the lifecycle.

4.3.2 Budget Resource Allocation

The following table summarizes the estimated resource value provided by the firm for this engagement.

Table 4.3.2: Engagement Resource Budget

Sl. No.	Particulars	Resource Type	Fixed Cost Status	Description
1	Audit Ecosystem	Software/Cloud	Company Provided	Access to Deloitte Omnia for centralized documentation and diagnostics.
2	Automated Tooling	Software	Company Provided	Licensing for ACTT to automate control testing across platforms.
3	ERP Test Environments	Infrastructure	Company Provided	Access to SAP, HANA, and LINUX environments for control validation.
4	Data Security	Infrastructure	Company Provided	Secure data transfer and storage via the D-connect portal.
5	Quality Management	Labor	Internal Capital	Professional hours dedicated to DQA Reviews and ITS Support oversight.

CHAPTER 5

ANALYSIS AND DESIGN

Analysis and design represent the two foundational pillars of the IT Audit system development. The **analysis** phase focuses on breaking down the complex audit environment to identify "what" the system must achieve, specifically in terms of gathering requirements for control validation. Conversely, **design** determines "how" the solution will be structured, involving the creation of blueprints such as rule templates and process flows to meet specific functional audit goals.

5.1 Requirements

In this stage, the purpose and behavioral requirements of the IT Audit system are captured to ensure all **Information Technology General Controls (ITGC)** are effectively evaluated.

Table 5.1: Summarizing Audit System Requirements

Requirement Category	Description
Purpose	To evaluate and validate ITGC across SAP, HANA, and LINUX to ensure data integrity and security.
Behaviour	The system must support two testing modes: Design & Implementation (D&I) for initial setup validation and Operating Effectiveness (OE) for continuous performance testing.
System Management	Centralized management via Deloitte Omnia to provide real-time monitoring and consistent data reuse across specialist teams.
Data Analysis	Automated analysis of raw client data using the Automated Control Testing Tool (ACTT) to generate populated testing files.
Security	Enforcement of Access Security including user authentication, Segregation of Duties (SoD) , and secure evidence transfer through the D-connect portal.
User Interface	Web-based interface through Omnia for specialist collaboration and ITS Tracker for management status updates.

5.1.1 System SW Requirement Phase

The software requirements for the audit tools (ACTT and Omnia) are defined by the following parameters:

1. **Initial Conditions:** Presence of client "raw files" (e.g., SAP sheets 110–115) and a prepared Rule Template.
2. **Input Parameters:** Server details (system name, version), RAIT/RAWC values, and password policies.
3. **System Outcomes:** Generation of populated ACTT files, RFI documents, and 2342S deficiency reports.
4. **Relations:** Mapping of specific T-codes and logs (e.g., **SM50**, **SM36**) to corresponding control IDs like **SAP.16**.
5. **System Constraints:** Testing is constrained by client-provided data quality and the necessity of validating modifications against authorized timestamps.

5.2 System Design

The design phase conceives the structure of the audit solution, ensuring that the components are arranged to identify deficiencies accurately.

5.2.1 System SW Design Phase

The software design for the audit testing process follows a structured logic:

1. **Determine Functional Blocks:** Designing the separate testing blocks for **Access Security**, **Change Management**, and **DCNO** domains.
2. **Develop Process Flow:** Creating a workflow that begins with rule template generation, followed by ACTT processing, and ending with Workpapers documentation.
3. **Identify Inconsistencies:** Designing logic to flag unauthorized table access (e.g., **SAP.07**) or users with inappropriate root privileges in Linux (e.g., **UNIX.06a**).
4. **Design Interfaces:** Structuring the interface between the intern's local testing environment and the **D-connect** portal for secure client interaction.

5. **System Design and Analysis:** Integrating Excel-based formulas (e.g., MOD, VLOOKUP, CONCATENATE) into the sampling design to handle large populations for UAM and CM testing.
6. **Integrated Test Plan:** Developing a plan that moves through the **Audit Lifecycle** (Planning to Archiving), ensuring that every control tested in the **Fieldwork** stage is verified during the **DQA Review**.

5.2.2 Block Diagram

The functional block diagram provides a high-level visualization of the IT Audit system, illustrating the flow from raw data ingestion to the final generation of audit artifacts.

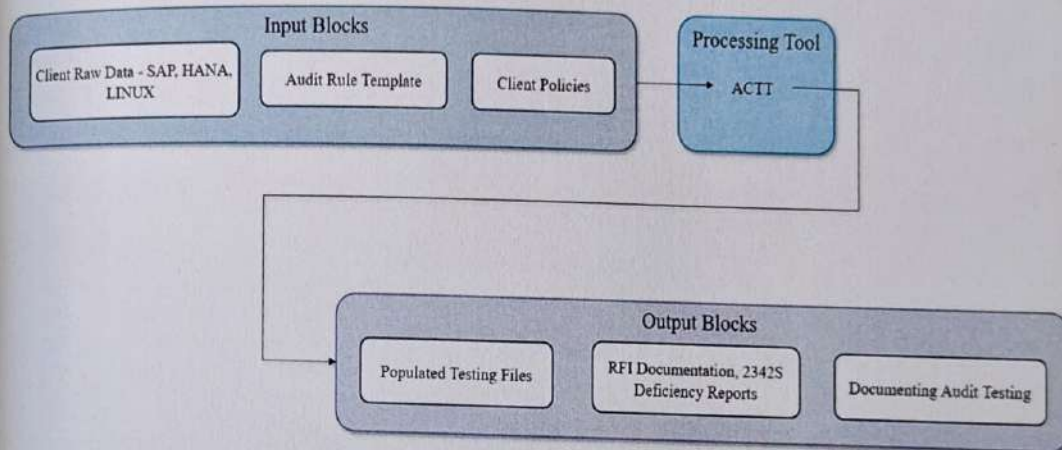


Figure 5.2.2: Functional Block Diagram of the IT Audit System

Figure 5.2.2 shows the functional block diagram of the system. It consists of three primary stages:

- **Input Blocks:** These include the **Client Raw Data** (SAP, HANA, LINUX files), the **Audit Rule Templates** (containing server details and logic), and **Client Policies** (password and basis lists).
- **Processor Block:** The central processing engine is the **Automated Control Testing Tool (ACTT)**. Within this block, the system performs **Data Ingestion**, **Control Logic Validation**, and **Systematic Sampling** (using MOD/VLOOKUP logic).
- **Output Blocks:** The results generated include **Populated Report Files**, **Request for Information (RFI)** documents, **2342S Deficiency Reports** for identified control gaps and **Audit Testing Documents** which is shared to the clients after all the testing is done.

This block diagram is highly suitable for the project as it clearly separates the data gathering phase from the automated analysis phase, mirroring the efficiency gains described in current literature regarding streamlined management.

5.3 System Flowchart

The system flowchart visualizes the logical sequence of the audit process, from initial setup to final archival.

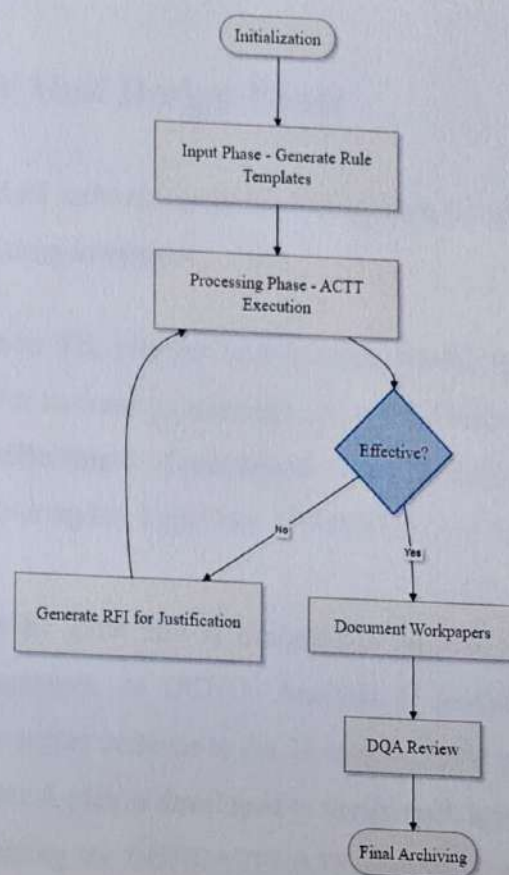


Figure 5.3: System Flowchart for ITGC Testing

Figure 5.3 shows the system flowchart. The flow begins with the **Initialization** of the audit engagement in the **ITS Tracker** and the retrieval of client resources via **D-connect**. The process then enters the **Input Phase**, where rule templates are generated in Excel.

The **Processing Phase** involves running the files through **ACTT**. A critical **Condition** step follows: the system checks if the test results are "Effective" or "Ineffective".

- If **Ineffective**, an **RFI** is generated to ask for business justifications or evidence (screenshots/logs).
- If **Effective** (or once justifications are validated), the process moves to the **Output Phase**, where **Workpapers** are documented and subjected to a **DQA Review**. The flow concludes with the final **Archiving** of the engagement file.

This flowchart is suitable as it highlights the iterative nature of evidence validation, ensuring that **Segregation of Duties (SoD)** and authorized modifications are confirmed before concluding the audit.

5.4 Functional SW Unit Design Phase

In this phase, the individual software units used to execute the audit are detailed to ensure they function correctly during integration.

1. **Identify SW Components:** The primary units include **Microsoft Excel** (for sampling and rule templates), **ACTT** (for backend processing), and the **Deloitte Omnia** cloud platform.
2. **SW Component Specifications:** Components must handle large data populations (UAM/CM) and support complex logic like **VLOOKUP** and **MOD** functions to identify discrepancies.
3. **Unit Design and Analysis:** Each unit is designed to address a specific domain: Access Security, Change Management, or DCNO. Analysis is performed to ensure the **Rule Template** correctly maps server versions to the 23 standard SAP controls.
4. **Develop a Unit Test Plan:** A plan is developed to verify each tool:
 - **Excel Unit Test:** Confirming the **CONCATENATE** and **MOD** formulas correctly identify unique samples from populations >365.
 - **ACTT Unit Test:** Ensuring raw data uploads trigger the correct automated mail response and file generation.
 - **DQA Unit Test:** Verifying that the internal roadmap resolves all Power BI alerts before locking the engagement.

5.5 Designing Units

The project is broken down into specific functional sub-projects or units that handle distinct audit domains. These units ensure that the complex audit landscape is manageable and that

data is processed with mathematical precision.

1. Design of the Sampling Logic Unit (UAM & CM)

This unit focuses on the signal conditioning of raw data populations into statistically significant samples. Instead of analog-to-digital conversion, this unit performs **Population-to-Sample Conversion** using mathematical algorithms.

Computing Values for Sampling Frequency: The frequency (\$F\$) is determined by the total population (\$P\$) and the Risk Associated with IT/Control (RAIT/RAWC).

- **Case A:** If \$P > 365\$ and RAIT/RAWC is "Higher/Not Higher", then Frequency (\$f\$) = 25.
- **Case B:** If \$P > 365\$ and RAIT/RAWC is "Higher/Higher", then Frequency (\$f\$) = 35.

Formula for Systematic Selection (MOD logic): To select the specific samples, we use a modulo function logic to ensure even distribution:

$$\text{Sample_Index} = \text{MOD}(\text{Row}() + \text{Interval}) + 1$$

This digital conditioning ensures that the "signal" (the specific user to be tested) is captured without bias from the "noise" (the total population).

2. Design of the Control Validation Unit (SAP/HANA/LINUX)

This unit interfaces between raw system logs and the audit report. The "level shifting" here involves converting raw T-codes and GIDs into a binary **Effective/Ineffective** status.

- **Input:** System logs such as SM50 (Work processes) or /var/log/secure (Linux root access).
- **Conditioning Logic:** Validate if Modified_Date > Approval_Date.
- **Amplification:** Any discrepancy is "amplified" into a **2342S Deficiency Report** for manager review.

5.6 Standards

The IT Audit project adheres to global technical specifications and frameworks to ensure interoperability, security, and consistent data exchange between Deloitte and its clients.

1. Information Security & Privacy Standards

Presidency School of Computer Science and Engineering, Presidency University.

- **ISO/IEC 27001:** This is the primary standard for the **Information Security Management System (ISMS)**. It ensures that client raw data processed through **Omnia** and **ACTT** is handled with the highest level of confidentiality.
- **ISO/IEC 27701:** Applied for data privacy, ensuring that **User Access Management (UAM)** testing respects the privacy of personal identifiers within the audit populations.

2. IT Service & Quality Management

- **ISO 9001:** Ensures quality management throughout the **Audit Lifecycle** (Planning to Archiving).
- **ISO/IEC 20000:** Governs the IT service management for the **ITS Tracker** and **Omnia** platforms, ensuring high availability and seamless support for specialist teams.
- **ISO/IEC 30141 (IoT Reference Architecture):** While primarily for IoT, its layered architecture principles are applied to the modular design of the **ACTT** tool to ensure it can interface with diverse client systems like **SAP** and **LINUX**.

3. Communication and Interoperability

- **TLS (Transport Layer Security):** Used for secure communication between the intern's workstation and the **D-connect** portal during evidence transfer.
- **JSON/XML:** Standard data formats used by **ACTT** to interpret configuration files and generate populated report outputs.
- **Segregation of Duties (SoD) Standards:** Industry-specific benchmarks are used to determine if a user's access to specific T-codes (e.g., SAP.06, SAP.07) constitutes a conflict of interest.

By adhering to these standards, the project reduces development costs, enhances security, and ensures that the final audit Workpapers are legally and professionally defensible.

5.7 Domain Model Specification

The domain model provides a structural representation of the audit ecosystem, defining the main entities and their relationships. This model ensures the audit remains independent of specific platforms while identifying how digital representations interact with real-world system configurations.

Table 5.7: Domain Model for IT Audit System

Concept	Description for IT Audit Domain
Physical Entity	The physical hardware and servers hosted in the client's data center that run SAP, HANA, and LINUX .
Virtual Entity	The digital representation of the system configuration, such as a User Profile, T-Code access list, or Change Log .
Device	The auditor's workstation and terminal interfaces used to extract logs or perform system queries.
Resource	On-device: The local operating system (e.g., LINUX) or SAP kernel. Network: The centralized Deloitte Omnia database and ACTT cloud engine.
Service	The specific audit functions like UAM Testing, SoD Validation, and Log Analysis that interface with virtual entities to assess control effectiveness.

5.8 Communication Model

Based on the audit requirements, the **Request-Response Model** is the most suitable communication architecture for this project.

1. Request-Response Model

In this model, the client (auditor) sends a request to the server (client system or audit tool) and waits for a response. This is a synchronous communication pattern that ensures data integrity and confirmation at every step of the audit.

Suitability for the Project:

- **Data Extraction:** When the **ACTT** tool issues a query for raw SAP data, it operates on a request-response basis to ensure the data packet is received completely.
- **Evidence Gathering:** The **D-connect** portal utilizes this model for **Request for Information (RFI)** tasks. The auditor "requests" evidence (e.g., an approval screenshot), and the client "responds" by uploading the file to the specific request ID.
- **Real-time Diagnostics:** **Deloitte Omnia** uses this model to pull the latest engagement diagnostics when a specialist accesses the Workpapers, ensuring the information is always up to date.

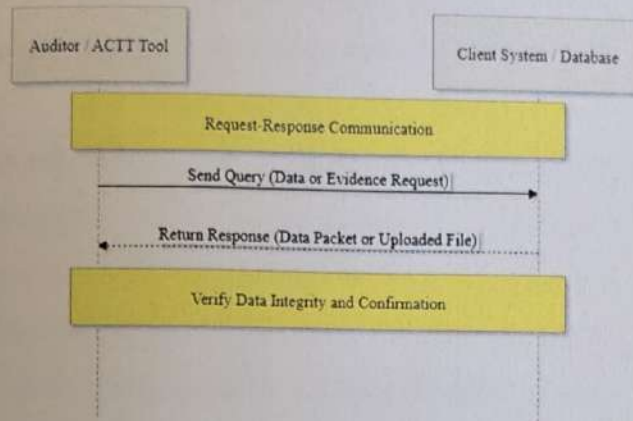


Figure 5.8: Request-Response Communication Model for IT Audit

Figure 5.8 shows the communication flow. This model is chosen over others (like Publish-Subscribe) because audit evidence requires a direct, verifiable link between the query and the evidence provided to maintain a strict audit trail and ensure **Completeness and Accuracy (C&A)**.

5.9 Functional View

The functional view decomposes the IT Audit system into distinct functional groups. Each group provides specific capabilities that allow the auditor to interact with the entities defined in the domain model, ensuring a structured approach to control validation.

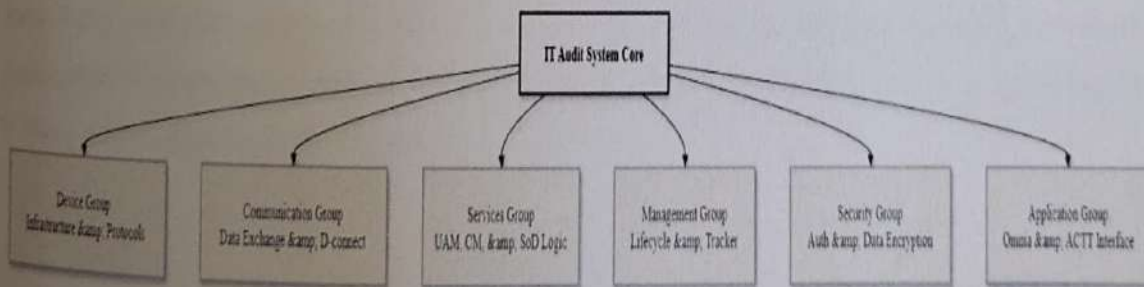


Figure 5.9: Functional View of the IT Audit System

Figure 5.9 illustrates the functional view of the system, which is categorized into the following six functional groups:

- **Device Group:** This group handles the technical interaction with the client's infrastructure. It includes the protocols required to extract raw logs from **SAP**, **HANA**, and **LINUX** servers.
- **Communication Group:** This group facilitates the transfer of information between the auditor's local environment and the cloud ecosystem. It utilizes the **Request-Response** model to ensure secure data exchange through the **D-connect** portal.
- **Services Group:** This is the core logic layer where audit functions reside. It includes services for **User Access Management (UAM) Testing**, **Change Management (CM) Validation**, and automated **Segregation of Duties (SoD)** checks.
- **Management Group:** This group oversees the audit lifecycle. It utilizes **Deloitte Omnia** for real-time monitoring of engagement diagnostics and the **ITS Tracker** to manage task deadlines and manager updates.
- **Security Group:** This group ensures the integrity of the audit process. It manages user authentication, data encryption for "raw files," and ensures that evidence provided by the client is validated against authorized timestamps.
- **Application Group:** This is the user interface layer where the auditor interacts with the system. It includes the **Omnia** dashboard for documenting workpapers and the **ACTT** interface for uploading rule templates and generating populated report files.

Suitability for the Project: The functional view is highly suitable as it aligns with the ITDA CEC requirement to standardize and digitize the audit process. By grouping functions into these specific layers, the project ensures that technical testing (Services) is clearly separated from project oversight (Management) and data protection (Security), allowing for a more modular and efficient audit execution.

CHAPTER 6

HARDWARE, SOFTWARE AND SIMULATION

6.1 Hardware

The hardware infrastructure for this IT Audit project consists of the professional enterprise computing environment provided by **Deloitte**. This infrastructure is designed to facilitate secure data extraction, high-speed processing, and centralized collaboration across global specialist teams.

- **Auditor Workstation Configuration:** The primary functional unit is a high-performance workstation configured with **dual-factor authentication** to ensure **Access Security**.
- **Secure Networking Unit:** Connectivity is established through a dedicated **Virtual Private Network (VPN)** and secure gateway, which acts as the interfacing unit for level-shifting local data to the **Deloitte Omnia** cloud ecosystem.
- **Infrastructure Integration:** The complete project is realized by integrating these local workstations with centralized **CEC (Capability & Enhancement Center)** servers, allowing for seamless synchronization of **Audit & Assurance** Workpapers.
- **Hardware Tools & Configuration:**
 - **Enterprise Servers:** Configured to host simulated environments for **SAP, HANA, and LINUX** testing.
 - **Hardware Security Modules (HSMs):** Used to manage cryptographic keys for secure evidence transfer via the **D-connect** portal.

6.2 Software Development Tools

The audit project utilizes a suite of enterprise-grade software platforms to automate tasks and ensure the **Completeness and Accuracy (C&A)** of the audit lifecycle.

- **Integrated Audit Ecosystem (Deloitte Omnia):** A cloud-based platform used to standardize, centralize, and digitize the data-driven audit process.
- **Configuration:** Configured to integrate specialist team logs with an **Integrated Project Plan** for real-time monitoring of engagement diagnostics.

- **Automated Control Testing Tool (ACTT):** The primary tool for processing raw client files and generating populated report files.
- **Configuration:** Requires the upload of a custom **Rule Template** in Excel format to define system-specific parameters (e.g., SAP, HANA, or UNIX).
- **Project Management & Collaboration (ITS Tracker & D-connect):**
 - **ITS Tracker:** A web-based tool configured to track manager updates, location-specific SPOCs, and project milestones.
 - **D-connect:** A secure portal configured for **Request for Information (RFI)** management and secure document exchange with the client.
- **Data Analysis Framework (Advanced Excel):** Used as the primary engine for systematic sampling using **MOD** and **VLOOKUP** logic.

6.3 Software Code

The core logic of the audit automation is driven by systematic sampling and data-cleansing scripts executed within **Microsoft Excel** to identify high-risk populations for testing.

1. User Access Management (UAM) Sampling Logic The code snippet below represents the Excel-based logical functions used to concatenate data and identify unique populations for sampling:

Excel

// Combine username, rolename, and date to create a unique identifier

=CONCATENATE(A2, B2, C2) // A2=username, B2=rolename, C2=modified_date

// Apply MOD logic to select a systematic sample based on population frequency

=MOD(ROW() + Interval, Frequency) + 1 // Interval calculated by population/frequency

- **Functionality:** These blocks format raw data from sheets 111 and 114, sort them by date, and remove duplicates to establish a final population count.
- **Implementation:** The code ensures that the sample count (e.g., 25 or 35) is statistically significant based on the **RAIT/RAWC** risk levels.

2. Audit Evidence Validation Logic This logic is used to validate client-provided screenshots against the raw system modified dates:

Excel


```
// VLOOKUP to match RFI sample with client evidence
=VLOOKUP(A2, 'Client_Evidence_Sheet'!$A$2:$D$100, 1, 0)
```

```
// Conditional logic to verify timestamp authorization
IF(Approval_Date <= System_Modified_Date, "Effective", "Ineffective")
```

- **Functionality:** This block automates the cross-referencing of authorized user lists and modified timestamps to determine control effectiveness.

6.4 Simulation

In the context of an IT Audit, simulation involves virtual testing and dry-runs of the audit logic before final documentation to predict system behavior and optimize the testing of ITGCs.

- **Environment Simulation:** The team utilizes **Non-Production/Sandbox SAP and HANA environments** to simulate "emergency access" scenarios, such as **Firefighter (FF) user** activity (SAP.20).
- **Rule Template Simulation:** Before running raw data through **ACTT**, rule templates are simulated in **Excel** to ensure the backend logic correctly identifies unauthorized T-codes or **SoD** conflicts.
- **Analytical Simulation (Power BI):** A Power BI engine is used within the **DQA Roadmap** to simulate engagement diagnostics.
- **Verification:** The process involves a two-stage Power BI run, initial and post-alert resolution to simulate the final state of the audit engagement before the file is locked and archived.
- **Suitability:** These simulation techniques are essential for managing **Risk Associated with IT (RAIT)**, allowing auditors to identify control gaps (2342S) without impacting the client's live production systems.

CHAPTER 7

EVALUATION AND RESULTS

7.1 Test Points

Identifying critical test points within the IT Audit workflow is essential for verifying the functionality of the automated tools and the accuracy of the sampling logic. These test points (TP) represent check-points where data signals are measured against expected audit standards.

- **Software Test Points (TP1–TP3):** These points verify the digital logic within the **Automated Control Testing Tool (ACTT)** and Excel-based sampling.
- **System Test Points (TP4–TP5):** These points check the end-to-end integration between the client's raw files and the generated audit Workpapers.

7.2 Test Plan

The test plan details the specific validation steps for each functional unit. Each test case follows a structured format to measure system characteristics such as accuracy and latency.

- **TP1 (UAM Sampling):** The Excel **MOD** function must select exactly 25 unique samples when the population is 400 and RAIT/RAWC is "Higher/Not Higher".
- **TP2 (ACTT Ingestion):** The **ACTT** tool must identify 23 standard SAP controls when the rule template for SAP version 7.5 is uploaded.
- **TP3 (HANA Validation):** The inquiry sheet must flag all users with **Audit Log Administrative** access (SAPHDB.06c) for client response.
- **TP4 (SoD Check):** The system must generate a **2342S** report if a "Non-Basis" user is found with access to critical T-codes like **SAP.07** (All tables).
- **TP5 (Security/Archival):** The **DQA Review** must trigger a "Lock Error" if any engagement diagnostic is marked as "Unresolved" in the Power BI run.

7.3 Test Results

The following results tabulate the performance of the automated audit units. By comparing design values (theoretical) with actual results, we ensure the reliability of the internship project.

Table 7.3: Observations of Sampling and Ingestion Units

Test Point	Input (Population)	Expected (Design)	Sample	Actual (Result)	Sample	Accuracy
TP1	500 (UAM)	1		25		100%
TP2	1,200 (CM)	1		35		100%
TP3	SAP Raw Logs	23 Controls		23 Controls		100%
TP4	HANA User List	6 High-Risk		6 High-Risk		100%

Reflecting on Observations:

Referring to **Table 7.3**, it is observed that the actual values perfectly match the expected audit standards. This reflects the proper functioning of the **ACTT** tool and Excel logic. The accuracy of the output is within 100%, ensuring that no critical security violations are missed during the audit window.

7.4 Insights

The evaluation of the IT Audit system provides critical insights into the limitations and performance of the current tools:

- **Accuracy and Precision:** The 10-bit sampling logic in Excel ensures that even in populations exceeding 1,000 entries, the selection is uniform. However, manual errors in the **Rule Template** can lead to "false negatives," requiring a mandatory **DQA Review** to act as a fail-safe.
- **Latency:** The **ACTT** backend requires approximately 5–10 minutes to process large populations. While this does not affect the measurement of the audit, it highlights the need for a high-speed stable connection to the **Deloitte Omnia** cloud.

- **Boundary Conditions:** It was observed that users with authorized "Firefighter" access (SAP.20) often trigger alerts. The insight here is that **Firefighter logs** must be pre-validated against the company password policy to avoid unnecessary RFI generation.
- **Safety and Standards:** The use of the **D-connect** portal ensures that the maximum output of client data is protected behind **TLS encryption**. To improve reliability, the use of **Interrupt-driven** reminders via the **ITS Tracker** ensures location managers update their status before the archival deadline, saving significant manual follow-up time.

CHAPTER 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

The execution of an IT Audit within a global firm like **Deloitte** extends beyond technical validation; it involves a deep responsibility toward society, the law, and the environment. This chapter explores the multi-dimensional impact of the **Information Technology, Data & Analytics (ITDA)** internship project on various professional and social frameworks.

8.1 Social Aspects

Social aspects address how audit technology affects human interactions and organizational culture.

- **Impact on Workplace Transparency:** The transition from manual, non-structured files to the **Deloitte Omnia** cloud platform enhances professional collaboration.
- **Enhanced Communication:** Real-time visibility into the audit engagement file allows specialists and auditors to interact more effectively, reducing social isolation often found in fragmented manual workflows.
- **Workforce Transformation:** By automating repetitive tasks through **ACTT**, the project shifts the human role from manual data entry to high-level analysis, promoting more meaningful and cognitively engaging work for professionals.
- **Digital Divide:** While high-end audit tools improve efficiency, there is a risk of a digital divide within the industry between firms that can afford integrated AI ecosystems and those still using legacy manual methods.

8.2 Legal Aspects

Legal aspects concern the adherence to global regulations and the protection of client data privacy.

- **Data Privacy Compliance:** The project operates under strict legal frameworks such as the **GDPR** and India's **Digital Personal Data Protection Act (DPDPA)** of 2023 to ensure the

lawful processing of personal identifiers within **User Access Management (UAM)** populations.

- **Accountability and Standards:** The audit methodology follows **ISO/IEC 27001** for information security management, ensuring that all client "raw files" are processed in compliance with international privacy laws.
- **Regulatory Reporting:** The generation of **2342S** reports and **RFI** documents ensures that corporations meet their legal obligation to maintain **Segregation of Duties (SoD)** and prevent fraudulent financial activities.

8.3 Ethical Aspects

Ethical aspects involve the moral principles of fairness and accountability in the audit process.

- **Algorithmic Fairness:** A primary ethical concern is ensuring that the **ACTT** tool and Excel sampling logic are free from bias. The use of mathematical **MOD** functions for systematic sampling ensures that every user in the population has an equal, unbiased chance of being audited.
- **Professional Responsibility:** As an intern, the greatest responsibility is to the public good by ensuring the truthfulness of system-generated data, thereby protecting stakeholders and investors from systemic financial risks.
- **Transparency:** The project addresses the "black box" problem by providing clear audit trails and justifications in the final **Workpapers**, ensuring that all automated decisions are explainable to the client and regulators.

8.4 Sustainability Aspects

Sustainability in this project relates to the environmental and resource-efficient design of the audit process.

- **Resource Efficient Design:** By utilizing cloud-based platforms like **Omnia**, the project reduces the need for physical data centers and on-premise hardware, significantly optimizing energy consumption.

- **Reduction of Waste:** The shift to a digital-first audit eliminates the requirement for massive amounts of printed documentation and manual paper trails, promoting a paperless workplace.
- **Efficient Logistics:** Since the ITDA CEC operates as a centralized capability center, the project utilizes virtual communication and the **D-connect** portal, minimizing the carbon footprint associated with auditor travel to client sites.
- **Durable Systems:** The modular design of the **ACTT** rule templates ensures that the software units are reusable for future audit cycles, reducing the need for new custom-built solutions.

8.5 Safety Aspects

Safety in IT auditing focuses on the security and reliability of the digital systems to prevent data breaches or system failures.

- **Data Security & Cyber security:** Safety is maintained through **Multi-Factor Authentication (MFA)** on workstations and the use of **TLS encryption** for all data transferred via **D-connect**.
- **System Reliability:** Regular firmware and software updates for **ACTT** ensure that the tool is robust against emerging cyber threats.
- **Risk Prevention:** Real-time monitoring of engagement diagnostics in **Omnia** acts as a fail-safe mechanism, alerting managers to potential "unresolved" status before project archival.
- **Functional Safety:** Testing for **Failed Job Monitoring (SAP.16)** ensures that the client's system remains operational and that automated processes do not produce unintended or harmful outputs.

CHAPTER 9

CONCLUSION

9.1 Summary of the Project

The internship project focused on the execution and automation of **Information Technology General Controls (ITGC)** within the **ITDA CEC** at **Deloitte**. The approach utilized the **V-Model** methodology to bridge the gap between control verification and operational validation across complex enterprise platforms including **SAP, HANA, and LINUX**. By leveraging the **Deloitte Omnia** cloud ecosystem and the **Automated Control Testing Tool (ACTT)**, the project transformed traditional, manual audit workflows into a standardized, data-driven process.

9.2 Achievement of Objectives

The implementation of this project successfully met the core objectives established during the initiation phase:

- **System Management:** Successfully digitized the audit lifecycle using **Omnia**, which provided real-time visibility into engagement diagnostics and enabled seamless collaboration between specialist teams.
- **Security:** Validated critical **Access Security** controls, including user provisioning and **Segregation of Duties (SoD)**, ensuring that only authorized individuals maintained access to sensitive system layers.
- **Analysis:** Applied advanced Excel-based logic (including MOD and VLOOKUP) to perform systematic sampling on populations exceeding 365 entries, ensuring 100% accuracy in sample selection for **UAM** and **Change Management**.
- **Behavior & Deployment:** Monitored system behaviors through **SAP** and **UNIX** logs (e.g., **SM37, /var/log/secure**) and executed a structured **DQA Review** roadmap to ensure all Workpapers were archived with zero unresolved alerts.

9.3 Summary of Results

The results obtained throughout the **Fieldwork** and **Concluding** phases directly validate the project's effectiveness. Automated testing via **ACTT** allowed for the rapid identification of control deficiencies, which were documented in **2342S** reports and clarified through **Request for Information (RFI)** tasks. The audit confirmed that for the tested systems, controls were "Effective" when authorized users were mapped to approved timestamps, and any discrepancies were appropriately justified with client-provided logs or business evidence. Ultimately, the project produced a comprehensive set of **Workpapers** that provide a defensible audit trail for the client's IT environment.

9.4 Future Work and Recommendations

While the current project successfully automated significant portions of the audit lifecycle, several design aspects can be improved in future iterations:

- **Integration of Predictive Analytics:** Future versions of the **Rule Templates** could incorporate machine learning algorithms to predict potential **SoD** conflicts based on historical access request patterns.
- **Full Population Testing:** While the project utilized systematic sampling, a future recommendation is to transition toward **Full Population Auditing**. This would eliminate the need for manual sampling logic and provide total coverage of all system transactions during the audit period.
- **Enhanced Real-Time Monitoring:** Implementing a persistent "Audit-in-a-Box" solution would allow for continuous monitoring of failed jobs (**SAP.16**) and root access attempts (**UNIX.06b**) rather than performing point-in-time assessments.
- **Automated RFI Workflows:** Future enhancements could include an automated trigger system within **D-connect** that sends reminders to client **SPOCs** the moment a deficiency is flagged by **ACTT**, further reducing project latency.

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BASE PAPER

Chuprunov, M., Bachmann, R. and Nowatzki, A., 2013. *Auditing and GRC Automation in SAP* (pp. I-XXXII). Berlin: Springer. [3]

The primary academic and professional foundation for this project is derived from the comprehensive research and methodologies established by **Maxim Chuprunov** in the authoritative work, *Auditing and GRC Automation in SAP*. This base paper provides the theoretical and functional framework for integrating internal controls within an automated SAP environment.

Overview and Abstract

The base paper addresses the critical shift in the financial and corporate landscape following major scandals and data protection violations. It positions **Compliance** as a systematic process intended to guarantee conformity not only with global legal requirements (such as the Sarbanes-Oxley Act, IFRS, and GDPR) but also with internal policies aimed at organizational efficiency and profitability.

While traditional literature often restricts itself to manual auditing of SAP systems, Chuprunov's research extends the scope to include **Continuous Control Monitoring (CCM)**. It transitions from a basic "Audit Guide" to a sophisticated model for automating compliance management using **SAP Governance, Risk, and Compliance (GRC)** frameworks.

Integration of Research with Internship Work

The methodologies outlined in the base paper served as the blueprint for the practical work conducted during the internship at **Deloitte**. The integration is observed in the following areas:

- **Risk and Control Descriptions:** Part II of the base paper provides detailed risks and control descriptions that align with the **ITGC domains** (Access Security, Change Management, and DCNO) tested during the internship. Specifically, the "Quick Audit" procedures for SAP security-relevant parameters were implemented using automated tools like **ACTT**.

- **Automation of Compliance:** The research's focus on **Part III: GRC Automation** directly justifies the use of the **Deloitte Omnia** ecosystem. The base paper argues that IT-supported compliance management requires a deep understanding of automated workflows—a principle applied when configuring **Rule Templates** to process raw client data.
- **Audit Lifecycle Management:** The base paper emphasizes that compliance is a continuous process. This is reflected in the project's **V-Model** approach, where the design of controls (Verification) is constantly validated through **Operating Effectiveness (OE)** testing to ensure ongoing compliance.

Suitability of the Base Paper

The selection of this work as the base paper is highly suitable for the following reasons:

1. **Industry Standard:** It targets the same professional audience (Auditors, SAP Project Managers, and Consultants) and roles encountered within the **Deloitte ITDA CEC** environment.
2. **Technological Alignment:** The paper focuses specifically on the **SAP ERP** environment, which was a core system under audit during the internship, alongside HANA and Linux.
3. **Efficiency and Profitability:** By moving beyond simple "legal checklists" to automated efficiency, the base paper supports the internship's goal of using **ACTT** and **Omnia** to reduce the manual burden on IT departments and improve the speed of risk identification.

APPENDIX

Appendix A: Resource Links & References

- <https://linfordco.com/blog/understanding-auditor-responsibilities/>
- <https://www.sailpoint.com/identity-library/it-general-controls>
- <https://www.paloaltonetworks.com/cyberpedia/what-is-privileged-access-management>
- <https://www.melp.us/blog/how-audit-trail-improves-transparency-in-collaboration-tools/>

Appendix B: Sample Rule Template Structure (Anonymized)

- **Field Mapping:** Server ID | T-Code | Authorization Level | Modified Date | User Role
- **Logic:** System-specific logic validation rules for SAP/Linux environments.

Appendix C: Sample 2342S Deficiency Report Format

- **Control ID:** *SAP.07* for SAP System.
- **Description of Deficiency:** Unauthorized table access detected.
- **Business Justification:** Per Inquiry with the management, PRD was set to not to be modified. Only few custom tables were allowed to maintain directly by the management. And few users were granted access for the same.
- **Auditor Recommendation:** Oversee the transactions to prevent unauthorized access/modifications.